

Section : Chemistry

Q.1 What is the structure of anionic coordination sphere in $[\text{Cr}(\text{NH}_3)_6][\text{CuCl}_5]$ complex?

1. Trigonal bipyramidal
2. Square pyramidal
3. Tetrahedral
4. Square planar

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Match List - I and List - II

List - I List - II

Name of vitamins Deficiency Disease

- | | |
|-----------------|------------------------------------|
| (A) Vitamin - A | (I) Increase in fragility of RBC |
| (B) Vitamin - E | (II) Increased blood clotting time |
| (C) Vitamin - C | (III) Xerophthalmia |
| (D) Vitamin - K | (IV) Scurvy |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.3 The correct statements for the activation energy of a reaction are :

- (A) Activation energy is the energy difference between the activated complex and the total energy of the reactants.
- (B) Higher is the activation energy, the faster is the rate of reaction.
- (C) Catalyst increases the rate of the reaction by following the path of lower activation energy.
- (D) In the Arrhenius equation, E_a corresponds to the activation energy

Choose the correct answer from the options given below :

1. (A), (B) and (D) only
2. (A) and (D) only
3. (A), (C) and (D) only
4. (A), (B), (C) and (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Which of the following statements with reference to a solution are correct?

- (A) the molarity of the solution changes with the change in the temperature.
- (B) the volume of the solution changes with the change in the temperature.
- (C) the molality of the solution changes with the change in the temperature.
- (D) the normality of the solution changes with the change in the temperature.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Select the correct increasing order of conductivity of the given complexes in their solution

- (A) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- (B) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
- (C) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
- (D) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (D), (B), (A)
3. (D), (C), (B), (A)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.6 The unit of rate constant of a reaction with the rate law, $\text{Rate} = k[\text{A}]^{3/2}[\text{B}]^{-1}$ is

1. s^{-1}
2. $\text{mol L}^{-1} s^{-1}$
3. $\text{L}^{1/2} \text{mol}^{-1/2} s^{-1}$
4. $\text{mol}^{1/2} \text{L}^{-1/2} s^{-1}$

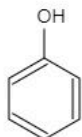
Options 1. 1

2. 2
3. 3
4. 4

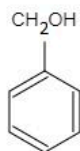
Q.7

Which of the following compounds are aromatic alcohols?

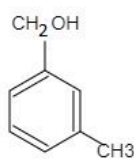
(A).



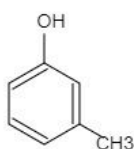
(B).



(C).



(D).



Choose the **correct** answer from the options given below:

1. (A), (B) and (D) only
2. (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.8 The colour of $[\text{Ni}(\text{NO}_2)_6]^{4-}$ complex is _____

1. Brown-yellow
2. Green
3. Blue
4. Violet

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Which of the following reagents will not convert Toluene into Benzaldehyde ?

1. CrO_2Cl_2
2. CrO_3
3. Cl_2 / light, H_2O
4. CO/HCl in presence of AlCl_3

Options

1. 1
2. 2
3. 3
4. 4

Q.10 What is the product formed when tertiary alcohol is treated with alkaline KMnO_4 at elevated temperature?

1. Ketone
2. Aldehyde
3. A mixture of carboxylic acids
4. Carboxylic acid

Options

1. 1
2. 2
3. 3
4. 4

Q.11 The chemical composition of Tollens' reagent is

1. Sodium bicarbonate
2. Mercurous chloride
3. Iodoform
4. Ammoniacal silver nitrate solution

Options

1. 1
2. 2
3. 3
4. 4

Q.12 Which of the following base is not present in DNA ?

1. Adenine
2. Guanine
3. Cytosine
4. Uracil

Options

1. 1
2. 2
3. 3
4. 4

Q.13 sp^3 hybridization leads to _____

1. tetrahedral geometry with bond angles 90° each
2. tetrahedral geometry with bond angles 109.5° each
3. trigonal geometry with bond angles 120° each
4. square planar geometry with bond angles 90° each

Options

1. 1
2. 2
3. 3
4. 4

Q.14 Which of the following complexes is heteroleptic?

1. $[\text{Fe}(\text{NH}_3)_4\text{Cl}_2]^+$
2. $[\text{Cr}(\text{NH}_3)_6]^{3+}$
3. $[\text{Co}(\text{NH}_3)_6]^{3+}$
4. $[\text{Mn}(\text{CN})_6]^{4-}$

Options

1. 1
2. 2
3. 3
4. 4

Q.15 Arrange the following compounds in increasing order of boiling point.

- (A) Ethanol
- (B) Diethyl ether
- (C) Ethanal
- (D) Ethane

Choose the correct answer from the options given below:

1. (A), (B), (D), (C)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (D), (B), (C), (A)

Options

1. 1
2. 2
3. 3
4. 4

Q.16 The composition of starch in amylopectin is

1. 15-20%
2. 20-40%
3. 50-70%
4. 80-85%

- Options 1. 1
2. 2
3. 3
4. 4

Q.17 What is the total number of structural isomers possible for the compound with molecular formula C_4H_9Br ?

1. 5
2. 4
3. 3
4. 2

- Options 1. 1
2. 2
3. 3
4. 4

Q.18 Which of the following does not involve Pt as catalyst?

- (A) contact process for making H_2SO_4 .
(B) Ziegler-Natta catalyst.
(C) Adams catalyst, used for reduction.
(D) catalyst to decompose $KClO_3$ to give O_2 .

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B) and (C) only
3. (A), (B) and (D) only
4. (A) and (D) only

- Options 1. 1
2. 2
3. 3
4. 4

Q.19 Match the lanthanoid property given in List-I with lanthanoid ion given in List-II

- | List-I | List-II |
|--------------------------------|-----------------|
| Lanthanoid property | Ion |
| (A) Strong oxidant | (I) Lu^{3+} |
| (B) Strong reducing agent | (II) La^{3+} |
| (C) $[Xe]f^{14}$ configuration | (III) Ce^{4+} |
| (D) largest ionic radii | (IV) Eu^{2+} |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

- Options 1. 1
2. 2
3. 3
4. 4

Q.20 Reaction of $C_6H_5CH_2Br$ with aqueous sodium hydroxide follows

1. SN^1 mechanism
2. SN^2 mechanism
3. Mechanism is dependent on temperature of the reaction
4. Depends on concentration of aqueous sodium hydroxide

- Options 1. 1
2. 2
3. 3
4. 4

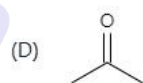
Q.21

Which of the following compound(s) does not react with Fehling solution?

(A) CH_3CHO

(B) Ph-CHO

(C) Glucose



Choose the **correct** answer from the options given below:

- (B) and (D) only
- (A), (B) and (C) only
- (A), (B), (C) and (D)
- (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.22 The correct expression for osmotic pressure of the solution involving dissociation of the solute is

1. $\pi = \frac{ic_2RT}{V}$

2. $\pi = ic_2RTV$

3. $\pi = \frac{n_2RT}{V}$

4. $\pi = \frac{in_2RT}{V}$

Options 1. 1

2. 2

3. 3

4. 4

Q.23 Arrange the following gases in the increasing order of their solubility in water at 298 K and atmospheric pressure.

(A) CH_4 ($K_H = 0.413 \text{ kbar}$)

(B) HCHO ($K_H = 1.83 \times 10^{-5} \text{ kbar}$)

(C) He ($K_H = 144.97 \text{ kbar}$)

(D) CO_2 ($K_H = 1.67 \text{ kbar}$)

Choose the correct answer from the options given below:

- (B), (A), (D), (C)
- (C), (D), (A), (B)
- (A), (B), (D), (C)
- (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.24 Match List-I with List-II

List-I

Electrolytes

(A) Non-electrolyte

(B) Binary electrolyte

(C) Quaternary electrolytes (III)

(D) Ternary electrolyte

List-II

Examples

(I) Sucrose, urea

(II) KCl , CH_3COOH H_2SO_4 BaCl_2

(IV)

 AlCl_3 K_3PO_4

Choose the correct answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3
4. 4

Q.25 Lanthanoids (Ln) when heated with nitrogen, form compounds of the type _____

1. Ln_2N
2. LnN_2
3. LnN
4. Ln_2N_3

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Match List-I with List-II

List-I	List-II
Compounds	Uses
(A) Dichloromethane	(I) Freon R 22
(B) Trichloromethane	(II) Synthesis of chlorofluorocarbons
(C) Triiodomethane	(III) Propellant
(D) Tetrachloromethane	(IV) Antiseptic

Choose the correct answer from the options given below :

1. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.27 The best reagent for preparing an alkyl chloride from an alcohol is

1. Concentrated HCl
2. SOCl_2
3. PCl_5
4. PCl_3

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Why must the reacting molecules have a minimum energy for an effective collision?

1. minimum energy is required for the particles to collide many times per second
2. to facilitate breaking of bonds and bond formation
3. the reaction mixture has sufficient energy to give off heat in a reaction
4. minimum energy is required to orient the particles correctly

Options 1. 1

2. 2
3. 3
4. 4

Q.29 The half life of a first order reaction having rate constant, $k = 1.7 \times 10^{-5} \text{ s}^{-1}$ is

1. 9.7 hrs
2. 11.3 hrs
3. 40.2 hrs
4. 18 hrs

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Formic acid does not respond to

1. Fehling's test
2. Tollen's test
3. sodium bicarbonate test
4. 2,4-DNP test

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Arrange the following compounds in increasing order of their boiling points.

- (A) Bromomethane (CH_3Br)
- (B) Bromoform (CHBr_3)
- (C) Chloromethane (CH_3Cl)
- (D) Dibromomethane (CH_2Br_2)

Choose the correct answer from the options given below:

1. (A), (C), (D), (B)
2. (C), (A), (D), (B)

3. (C), (A), (B), (D)

4. (C), (B), (A), (D)

Options 1. 1
2. 2
3. 3
4. 4

Q.32 Which of the following statements is incorrect?

1. When white light falls on the complex, some part of it is absorbed
2. Higher the crystal field splitting, lower will be the wavelength absorbed by the complex
3. Observed color of the complex is the color of the absorbed light/wavelength
4. In the absence of ligand, compound is colourless

Options 1. 1
2. 2
3. 3
4. 4

Q.33 What is the IUPAC name of m-cresol?

1. 3-chlorophenol
2. 3-methylphenol
3. 3-methoxyphenol
4. Benzene-1,3-diol

Options 1. 1
2. 2
3. 3
4. 4

Q.34 Select the colligative property which is used in the calculation of molar mass of biomolecules.

1. Depression in freezing point
2. Elevation in boiling point
3. Osmotic pressure
4. Relative lowering of vapour pressure

Options 1. 1
2. 2
3. 3
4. 4

Q.35 Match list-I and list-II

List-I

List-II

Common name IUPAC Name

- (A) oxalic acid (I) Ethanedioic acid
(B) Adipic acid (II) Alpha - hydroxy propanoic acid
(C) Lactic acid (III) Hexanedioic acid
(D) Phthalic acid (IV) Benzene-1,2-dicarboxylic acid

Choose the correct answer from the options given below :

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1
2. 2
3. 3
4. 4

Q.36 Arrange the following alkyl halides in order of increasing reactivity towards S_N1 reaction.

- (A) $(CH_3)_3C-I$
(B) $(CH_3)_3C-F$
(C) $(CH_3)_3C-Br$
(D) $(CH_3)_3C-Cl$

Choose the correct answer from the options given below:

1. (B), (D), (C), (A)
2. (A), (B), (C), (D)
3. (A), (C), (D), (B)
4. (D), (C), (A), (B)

Options 1. 1
2. 2
3. 3
4. 4

Q.37 Identify the incorrect statement for glucose.

1. Responds to Tollen's Test
2. On reduction with HI it forms n - hexane
3. Respond to Schiff's test
4. Respond to Fehling test

Options 1. 1
2. 2
3. 3
4. 4

Q.38 Anisole on heating with concentrated HI gives:

1. Iodobenzene
2. Iodobenzene and methanol
3. Phenol and methyl iodide
4. Phenol and methanol

Options 1. 1

2. 2
3. 3
4. 4

Q.39 The reaction, where all the species involved are in same phase, is called

1. Heterogeneous reaction
2. Bimolecular reaction
3. Homogeneous reaction
4. Elementary reaction

Options 1. 1

2. 2
3. 3
4. 4

Q.40 What is the oxidation state of Pt in $(\text{PtF}_5)_4$?

1. +3
2. +5
3. +4
4. +2

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the passage carefully and answer

When a metal is placed in a solution of its ions, the metal can lose electron and can go in the solution as ion or the metal ion in the solution can take electron from the electrode and get deposited as metal. Thus, the electrode acquires either positive or negative charge with respect to the solution leading to the development of a potential difference between the metal electrode and the solution. This potential difference is called the electrode potential of the metal. The electrode potential of a metal is a measure of relative tendency to undergo oxidation (loss of electron) or reduction (gain of electron). When the concentration of the ions in the solution is unity, the electrode potential is called standard electrode potential. The magnitude of potential depends upon the nature of electrode, concentration of ions in solution and the temperature. In a galvanic cell, the electrode at which oxidation occurs is known as anodic half-cell and the electrode at which reduction occurs is known as cathodic half cell. The emf of the cell in terms of standard reduction potential is given by $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$. The standard free energy change of the redox reaction taking place in the cell is related to E°_{cell} by $\Delta G^\circ = -nFE^\circ$. A redox reaction would occur spontaneously if the free energy change is negative. The equilibrium constant of the reaction is related to the standard free energy change by $\Delta G^\circ = -RT \ln K$.

The half cell reaction is the one that

1. Consumes half a mole of electrolyte
2. Takes place at one electrode of a Galvanic cell
3. Reduces the concentration of electrolyte to half of its initial value
4. Reduces the electrode potential to half its initial value

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Read the passage carefully and answer

When a metal is placed in a solution of its ions, the metal can lose electron and can go in the solution as ion or the metal ion in the solution can take electron from the electrode and get deposited as metal. Thus, the electrode acquires either positive or negative charge with respect to the solution leading to the development of a potential difference between the metal electrode and the solution. This potential difference is called the electrode potential of the metal. The electrode potential of a metal is a measure of relative tendency to undergo oxidation (loss of electron) or reduction (gain of electron). When the concentration of the ions in the solution is unity, the electrode potential is called standard electrode potential. The magnitude of potential depends upon the nature of electrode, concentration of ions in solution and the temperature. In a galvanic cell, the electrode at which oxidation occurs is known as anodic half-cell and the electrode at which reduction occurs is known as cathodic half cell. The emf of the cell in terms of standard reduction potential is given by $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$. The standard free energy change of the redox reaction taking place in the cell is related to E°_{cell} by $\Delta G^\circ = -nFE^\circ$. A redox reaction would occur spontaneously if the free energy change is negative. The equilibrium constant of the reaction is related to the standard free energy change by $\Delta G^\circ = -RT \ln K$.

The standard reduction potential of Zinc is represented by the reaction

1. $\text{Zn}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Zn}(\text{s})$
2. $\text{Zn}(\text{s}) \rightarrow \text{Zn}^{2+}(\text{aq}) + e^-$
3. $\text{Zn}^{2+}(\text{g}) + 2e^- \rightarrow \text{Zn}(\text{s})$
4. $\text{Zn}^{2+}(\text{s}) + 2e^- \rightarrow \text{Zn}(\text{s})$

Options 1. 1

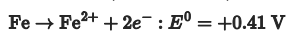
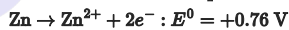
2. 2
3. 3
4. 4

Q.43 Read the passage carefully and answer

When a metal is placed in a solution of its ions, the metal can lose electron and can go in the solution as ion or the metal ion

in the solution can take electron from the electrode and get deposited as metal. Thus, the electrode acquires either positive or negative charge with respect to the solution leading to the development of a potential difference between the metal electrode and the solution. This potential difference is called the electrode potential of the metal. The electrode potential of a metal is a measure of relative tendency to undergo oxidation (loss of electron) or reduction (gain of electron). When the concentration of the ions in the solution is unity, the electrode potential is called standard electrode potential. The magnitude of potential depends upon the nature of electrode, concentration of ions in solution and the temperature. In a galvanic cell, the electrode at which oxidation occurs is known as anodic half-cell and the electrode at which reduction occurs is known as cathodic half cell. The emf of the cell in terms of standard reduction potential is given by $E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$. The standard free energy change of the redox reaction taking place in the cell is related to E_{cell}° by $\Delta G^{\circ} = -nFE^{\circ}$. A redox reaction would occur spontaneously if the free energy change is negative. The equilibrium constant of the reaction is related to the standard free energy change by $\Delta G^{\circ} = -RT \ln K$.

The standard oxidation potentials E° for the half cell reactions are



The emf of cell $\text{Fe}^{2+} + \text{Zn} \rightarrow \text{Zn}^{2+} + \text{Fe}$ is

1. -0.35 V
2. $+0.35 \text{ V}$
3. $+1.17 \text{ V}$
4. -1.17 V

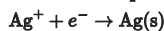
Options 1. 1

2. 2
3. 3
4. 4

Q.44 Read the passage carefully and answer

When a metal is placed in a solution of its ions, the metal can lose electron and can go in the solution as ion or the metal ion in the solution can take electron from the electrode and get deposited as metal. Thus, the electrode acquires either positive or negative charge with respect to the solution leading to the development of a potential difference between the metal electrode and the solution. This potential difference is called the electrode potential of the metal. The electrode potential of a metal is a measure of relative tendency to undergo oxidation (loss of electron) or reduction (gain of electron). When the concentration of the ions in the solution is unity, the electrode potential is called standard electrode potential. The magnitude of potential depends upon the nature of electrode, concentration of ions in solution and the temperature. In a galvanic cell, the electrode at which oxidation occurs is known as anodic half-cell and the electrode at which reduction occurs is known as cathodic half cell. The emf of the cell in terms of standard reduction potential is given by $E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$. The standard free energy change of the redox reaction taking place in the cell is related to E_{cell}° by $\Delta G^{\circ} = -nFE^{\circ}$. A redox reaction would occur spontaneously if the free energy change is negative. The equilibrium constant of the reaction is related to the standard free energy change by $\Delta G^{\circ} = -RT \ln K$.

The measured potential for



does not change with

1. Changing the temperature
2. Varying the concentration of Ag^{+} ion
3. Purity of Ag plate
4. Changing the pressure of the solution

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the passage carefully and answer

When a metal is placed in a solution of its ions, the metal can lose electron and can go in the solution as ion or the metal ion in the solution can take electron from the electrode and get deposited as metal. Thus, the electrode acquires either positive or negative charge with respect to the solution leading to the development of a potential difference between the metal electrode and the solution. This potential difference is called the electrode potential of the metal. The electrode potential of a metal is a measure of relative tendency to undergo oxidation (loss of electron) or reduction (gain of electron). When the concentration of the ions in the solution is unity, the electrode potential is called standard electrode potential. The magnitude of potential depends upon the nature of electrode, concentration of ions in solution and the temperature. In a galvanic cell, the electrode at which oxidation occurs is known as anodic half-cell and the electrode at which reduction occurs is known as cathodic half cell. The emf of the cell in terms of standard reduction potential is given by $E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$. The standard free energy change of the redox reaction taking place in the cell is related to E_{cell}° by $\Delta G^{\circ} = -nFE^{\circ}$. A redox reaction would occur spontaneously if the free energy change is negative. The equilibrium constant of the reaction is related to the standard free energy change by $\Delta G^{\circ} = -RT \ln K$.

The equilibrium constant of a feasible cell reaction is

1. Less than one
2. Greater than one
3. Equal to one
4. Zero

Options 1. 1

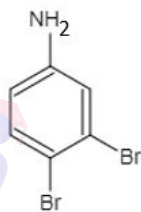
2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the questions

Amines are considered as derivatives of ammonia obtained by replacement of hydrogen atoms with alkyl or aryl groups. In nature they occur among proteins, vitamins, alkaloids and hormones. These can be primary, secondary or tertiary depending on the number of hydrogen groups replaced. They can be prepared from nitro compounds, alkyl halides, nitriles, amides etc. Due to unshared pair of electrons they behave as Lewis bases. Influence of the number of hydrogen atoms at nitrogen atom is displayed in physical properties such as solubility, boiling point and chemical reactions including basicity. Basic character of amines is expressed in terms of their K_b and pK_b values. Three classes of amines can be distinguished with p-toluene sulphonyl chloride which reacts with primary and secondary

amines. Amino group when attached to benzene ring behaves as an activating and ortho para directing group.

Give the IUPAC name of the compound



- 1,2-Dibromo-4-amino benzene
- 1,2-Dibromoaniline
- 1,2-Dibromoaminobenzene
- 3,4-Dibromoaniline

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the questions

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Which of the following is a tertiary amine ?

1. Trimethylamine
2. Allylmethylamine
3. Ethylmethylamine
4. Isopropylamine

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the questions

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Which reagent is used for the conversion of acetamide into ethylamine?

1. H_2/Ni
2. $LiAlH_4/H_2O$
3. Na/Ethanol
4. $LiAlH_4$ /ether

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the questions

Amines are considered as derivatives of ammonia obtained by replacement of hydrogen atoms with alkyl or aryl groups. In nature they occur among proteins, vitamins, alkaloids and hormones. These can be primary, secondary or tertiary depending on the number of hydrogen groups replaced. They can be prepared from nitro compounds, alkyl halides, nitriles, amides etc. Due to unshared pair of electrons they behave as Lewis bases. Influence of the number of hydrogen atoms at nitrogen atom is displayed in physical properties such as solubility, boiling point and chemical reactions including basicity. Basic character of amines is expressed in terms of their K_b and pK_b values. Three classes of amines can be distinguished with p-toluene sulphonyl chloride which reacts with primary and secondary amines. Amino group when attached to benzene ring behaves as an activating and ortho para directing group.

Which of the following reaction is not given by aniline ?

1. Alkylation
2. Acylation
3. Bromination
4. Friedel Craft's Reaction

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the questions

Amines are considered as derivatives of ammonia obtained by replacement of hydrogen atoms with alkyl or aryl groups. In nature they occur among proteins, vitamins, alkaloids and hormones. These can be primary, secondary or tertiary depending on the number of hydrogen groups replaced. They can be prepared from nitro compounds, alkyl halides, nitriles, amides etc. Due to unshared pair of electrons they behave as Lewis bases. Influence of the number of hydrogen atoms at nitrogen atom is displayed in physical properties such as solubility, boiling point and chemical reactions including basicity. Basic character of amines is expressed in terms

of their K_b and pK_b values. Three classes of amines can be distinguished with p-toluene sulphonyl chloride which reacts with primary and secondary amines. Amino group when attached to benzene ring behaves as an activating and ortho para directing group.

Which amine will have **highest pK_b** value?

1. $\text{CH}_3\text{CH}_2\text{NH}_2$
2. NH_3
3. $\text{C}_6\text{H}_5\text{NH}_2$
4. CH_3NH_2

Options 1. 1

2. 2

3. 3

4. 4